

第十二章 格林函数法

本章基本要求

1. 了解格林函数的意义;
2. 了解泊松方程的格林函数法求解方法;
3. 掌握用电像法求格林函数的方法, 及用格林函数法求解拉普拉斯方程的方法;
4. 了解含时间的格林函数, 掌握用冲量定理法求含时间的格林函数的方法。

§ 12.2 用电像法求格林函数

1. 在圆 $\rho = a$ 内求解拉普拉斯方程的第一边值问题

$$\begin{cases} \Delta_2 u = 0 \quad (\rho < a), \\ u|_{\rho=a} = f(\varphi). \end{cases} \quad (1)$$

此题即教材本节例 3。

解: 圆内第一边值问题的格林函数是

$$G(\mathbf{r}, \mathbf{r}_0) = -\frac{1}{2\pi} \ln \frac{1}{|\mathbf{r} - \mathbf{r}_0|} + \frac{1}{2\pi} \ln \frac{1}{|\mathbf{r} - \mathbf{r}_1|} + \frac{1}{2\pi} \ln \frac{a}{r_0},$$

其中, $\mathbf{r}$, $\mathbf{r}_0$ 和 $\mathbf{r}_1$ 分别是点 $M(\rho, \varphi)$, $M_0(\rho_0, \varphi_0)$ 和 $M_1(\rho_1, \varphi_0)$ 的径矢, M_1 是 M_0 关于圆 O 的电像 (图 12-1)。从而有

$$\rho_1 = a^2 / \rho_0$$

由图 12-1 可见

$$|\mathbf{r} - \mathbf{r}_0| = \sqrt{\rho^2 - 2\rho\rho_0 \cos(\varphi - \varphi_0) + \rho_0^2},$$

$$|\mathbf{r} - \mathbf{r}_1| = \sqrt{\rho^2 - 2\rho\rho_1 \cos(\varphi - \varphi_0) + \rho_1^2},$$

于是

$$G(\mathbf{r}, \mathbf{r}_0) = \frac{1}{2\pi} \left[\ln \sqrt{\rho^2 - 2\rho\rho_0 \cos(\varphi - \varphi_0) + \rho_0^2} - \ln \sqrt{\rho^2 - 2\rho\rho_1 \cos(\varphi - \varphi_0) + \rho_1^2} + \ln \frac{a}{\rho_0} \right].$$

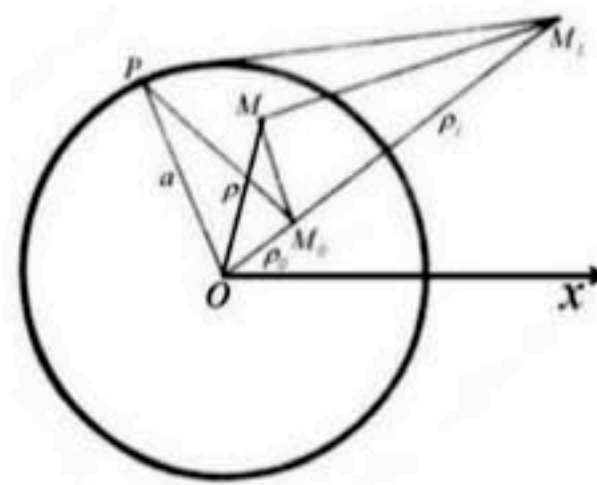


图12-1

$$\begin{aligned}
\left. \frac{\partial G}{\partial n_0} \right|_{\Sigma_0} &= \left. \frac{\partial G}{\partial \rho_0} \right|_{\rho_0=a} = \frac{1}{2\pi} \left[\frac{\rho_0 - \rho \cos(\varphi - \varphi_0)}{\rho_0^2 - 2\rho_0\rho \cos(\varphi - \varphi_0) + \rho^2} \right. \\
&\quad \left. - \frac{\rho_0 - \rho_1 \cos(\varphi - \varphi_0)}{\rho_0^2 - 2\rho_0\rho_1 \cos(\varphi - \varphi_0) + \rho_1^2} \right]_{\rho_0=a} \\
&= \frac{1}{2\pi} \left[\frac{a - \rho \cos(\varphi - \varphi_0)}{a^2 - 2a\rho \cos(\varphi - \varphi_0) + \rho^2} \right. \\
&\quad \left. - \frac{a - (a^2/\rho) \cos(\varphi - \varphi_0)}{a^2 - 2a(a^2/\rho) \cos(\varphi - \varphi_0) + a^4/\rho^2} \right] \\
&= \frac{[a - \rho \cos(\varphi - \varphi_0)] - (\rho/a)^2 [a - (a^2/\rho) \cos(\varphi - \varphi_0)]}{a^2 - 2a\rho \cos(\varphi - \varphi_0) + \rho^2} \\
&= \frac{1}{2\pi a} \frac{a^2 - \rho^2}{a^2 - 2a\rho \cos(\varphi - \varphi_0) + \rho^2}, \tag{2}
\end{aligned}$$

又

$$G(\mathbf{r}, \mathbf{r}_0)|_{\Sigma_0} = 0, \quad ds_0 = a d\varphi_0, \tag{3}$$

将②③代入解的积分公式(12.1.19), 得

$$\begin{aligned}
u(\rho, \varphi) &= \frac{1}{2\pi a} \int_0^{2\pi} \frac{(a^2 - \rho^2)f(\varphi_0)}{a^2 - 2a\rho \cos(\varphi - \varphi_0) + \rho^2} a d\varphi_0 \\
&= \frac{a^2 - \rho^2}{2\pi} \int_0^{2\pi} \frac{f(\varphi_0)}{a^2 - 2a\rho \cos(\varphi - \varphi_0) + \rho^2} d\varphi_0
\end{aligned}$$

此式即教材(12.2.22)式。

2. 在半平面 $y > 0$ 内求解拉普拉斯方程的第一边问题

$$\begin{cases} \Delta_2 u = 0 & (y > 0), \\ u|_{y=0} = f(x). \end{cases} \quad (1)$$

此题即教材本节例 4。

解：半平面第一边值问题的格林函数满足定解问题

$$\begin{cases} \Delta_2 G(x, y; x_0, y_0) = \delta(x - x_0)\delta(y - y_0), & (y > 0) \\ G|_{y=0} = 0. \end{cases}$$

其解为

$$G(\mathbf{r}, \mathbf{r}_0) = -\frac{1}{2\pi} \ln \frac{1}{|\mathbf{r} - \mathbf{r}_0|} + \frac{1}{2\pi} \ln \frac{1}{|\mathbf{r} - \mathbf{r}_1|},$$

其中, $\mathbf{r}$, $\mathbf{r}_0$ 和 $\mathbf{r}_1$ 分别是点 $M(x, y)$, $M_0(x_0, y_0)$ 和 $M_1(x_0, -y_0)$ 的矢径, M_1 是 M_0 关于平面 $y = 0$ 的电像 (图 12-2)。由图 12-2 可知

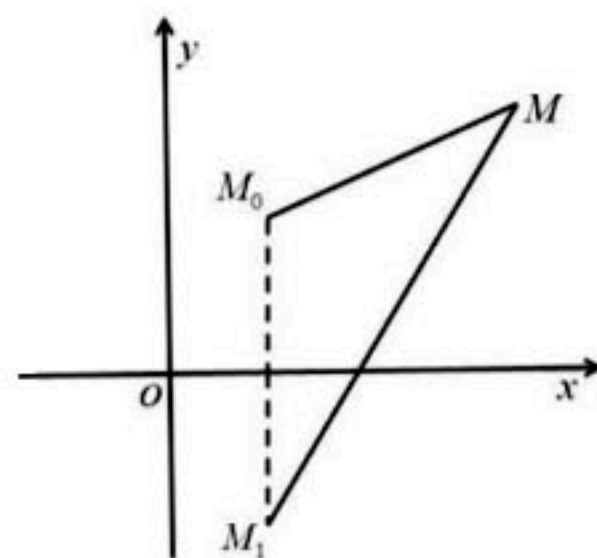


图12-2

$$\begin{aligned} |\mathbf{r} - \mathbf{r}_0| &= \sqrt{(x - x_0)^2 + (y - y_0)^2}, \\ |\mathbf{r} - \mathbf{r}_1| &= \sqrt{(x - x_0)^2 + (y + y_0)^2}, \end{aligned}$$

从而有

$$\begin{aligned} \left. \frac{\partial G}{\partial n_0} \right|_{\Sigma_0} &= - \left. \frac{\partial G}{\partial y_0} \right|_{y_0=0} = - \frac{1}{2\pi} \left[\frac{y_0 - y}{(x - x_0)^2 + (y - y_0)^2} \right. \\ &\quad \left. - \frac{y_0 + y}{(x - x_0)^2 + (y + y_0)^2} \right]_{y_0=0} \\ &= \frac{1}{\pi} \frac{y}{(x - x_0)^2 + y^2}, \end{aligned} \quad (2)$$

又

$$G(\mathbf{r}, \mathbf{r}_0)|_{\Sigma_0} = 0, \quad ds_0 = dx_0, \quad (3)$$

将 (2) (3) 代入解的积分公式 (12.1.19), 得

$$u(x, y) = \frac{y}{\pi} \int_{-\infty}^{\infty} \frac{1}{(x - x_0)^2 + y^2} f(x_0) dx_0.$$

此式即教材 (12.2.23) 式。

3. 在圆形域 $\rho < a$ 上求解 $\Delta u = 0$ 使满足边界条件 (1) $u|_{\rho=a} = A \cos \varphi$, (2) $u|_{\rho=a} = A + B \sin \varphi$, 本题如用 (12.2.22) 式, 则积分可利用留数定理算出, 参看 § 4.2 例 2 和习题 1(6)。本题亦即 § 8.1 习题 16, 可对照。

解:

$$(1) \quad \begin{cases} \Delta u = 0, & (\rho \leq a) \\ u|_{\rho=a} = A \cos \varphi, \end{cases} \quad (1)$$

由习题 1 的结果, (1) 的解为

$$u(\rho, \varphi) = \frac{a^2 - \rho^2}{2\pi} \int_0^{2\pi} \frac{A \cos \varphi_0}{a^2 - 2a\rho \cos(\varphi_0 - \varphi) + \rho^2} d\varphi_0 \quad (2)$$

由于是圆内问题, 有 $\rho < a$, (2) 式分子分母同除以 a^2 , 并令 $\frac{\rho}{a} = \varepsilon$, ($0 < \varepsilon < 1$), 有

$$u(\rho, \varphi) = \frac{(1 - \varepsilon^2)}{2\pi} \int_0^{2\pi} \frac{A \cos \varphi_0}{1 - 2\varepsilon \cos(\varphi_0 - \varphi) + \varepsilon^2} d\varphi_0 \quad (3)$$

作变换 $\cos \varphi_0 = \frac{1}{2}(z + z^{-1})$, $d\varphi_0 = \frac{1}{iz} dz$,

$$\begin{aligned} \cos(\varphi_0 - \varphi) &= \frac{1}{2}(e^{i\varphi_0} e^{-i\varphi} + e^{-i\varphi_0} e^{i\varphi}) \\ &= \frac{1}{2}(ze^{-i\varphi} + z^{-1}e^{i\varphi}), \end{aligned}$$

将这些变换关系代入 (3) 式, 有

$$\begin{aligned} I &= \int_0^{2\pi} \frac{\cos \varphi_0}{1 - 2\varepsilon \cos(\varphi_0 - \varphi) + \varepsilon^2} d\varphi_0 \\ &= -\frac{1}{2i} \oint_{|z|=1} \frac{(1 + z^2) dz}{z [\varepsilon e^{-i\varphi} z^2 - (1 + \varepsilon^2)z + \varepsilon e^{i\varphi}]} \end{aligned}$$

上式被积函数有三个单极点, 0 , $\varepsilon e^{i\varphi}$ 和 $e^{i\varphi}/\varepsilon$ 。其中, 0 和 $\varepsilon e^{i\varphi}$ 在 $|z|=1$ 单位圆内。其留数分别为 $e^{-i\varphi}/\varepsilon$ 和 $\frac{1 + \varepsilon^2 e^{i2\varphi}}{\varepsilon e^{i\varphi}(\varepsilon^2 - 1)}$, 留数之和为

$$\begin{aligned} &\frac{1}{\varepsilon} e^{-i\varphi} + \frac{1 + \varepsilon^2 e^{i2\varphi}}{\varepsilon e^{i\varphi}(\varepsilon^2 - 1)} \\ &= \frac{(1 + \varepsilon^2 e^{i2\varphi}) + (\varepsilon^2 - 1)}{\varepsilon e^{i\varphi}(\varepsilon^2 - 1)} = \frac{2\varepsilon}{(\varepsilon^2 - 1)} \cos \varphi \end{aligned}$$

则

$$I = -\frac{1}{2i} \cdot 2\pi i \cdot \frac{2\varepsilon}{(\varepsilon^2 - 1)} \cos \varphi = \frac{2\pi\varepsilon}{(\varepsilon^2 - 1)} \cos \varphi,$$

故

$$\begin{aligned} u(\rho, \varphi) &= \frac{1-(\rho/a)^2}{2\pi} \cdot A \cdot \frac{2\pi\rho/a}{1-(\rho/a)^2} \cos \varphi \\ &= \frac{A\rho}{a} \cos \varphi. \end{aligned}$$

$$(2) \begin{cases} \Delta u = 0, & (\rho < a) \\ u|_{\rho=a} = A + B \sin \varphi, \end{cases} \quad (4)$$

由习题 1 的结果, (4) 的解为

$$u(\rho, \varphi) = \frac{a^2 - \rho^2}{2\pi} \int_0^{2\pi} \frac{A + B \sin \varphi_0}{a^2 - 2a\rho \cos(\varphi_0 - \varphi) + \rho^2} d\varphi_0 \quad (5)$$

由于是圆内问题, 有 $\rho < a$, (5) 式分子分母同除以 a^2 , 并令 $\frac{\rho}{a} = \varepsilon$, ($0 < \varepsilon < 1$), 有

$$u(\rho, \varphi) = \frac{(1 - \varepsilon^2)}{2\pi} \int_0^{2\pi} \frac{A + B \sin \varphi_0}{1 - 2\varepsilon \cos(\varphi_0 - \varphi) + \varepsilon^2} d\varphi_0 \quad (6)$$

作变换 $\cos \varphi_0 = \frac{1}{2}(z + z^{-1})$, $\sin \varphi_0 = \frac{1}{2i}(z - z^{-1})$, $d\varphi_0 = \frac{1}{iz} dz$,

$$\begin{aligned} \cos(\varphi_0 - \varphi) &= \frac{1}{2}(e^{i\varphi_0} e^{-i\varphi} + e^{-i\varphi_0} e^{i\varphi}) \\ &= \frac{1}{2}(ze^{-i\varphi} + z^{-1}e^{i\varphi}), \end{aligned}$$

将这些变换关系代入 (6) 式, 有

$$\begin{aligned} I &= \int_0^{2\pi} \frac{A + B \sin \varphi_0}{1 - 2\varepsilon \cos(\varphi_0 - \varphi) + \varepsilon^2} d\varphi_0 \\ &= -\frac{1}{2i} \oint_{|z|=1} \frac{Bz^2 + 2Aiz - B}{z[\varepsilon e^{-i\varphi} z^2 - (1 + \varepsilon^2)z + \varepsilon e^{i\varphi}]} dz \end{aligned}$$

上式被积函数有三个单极点, 0 , $\varepsilon e^{i\varphi}$ 和 $e^{i\varphi}/\varepsilon$ 。其中, 0 和 $\varepsilon e^{i\varphi}$ 在 $|z|=1$ 单位圆内。其留数分

别为 $-\frac{B}{\varepsilon e^{i\varphi}}$ 和 $\frac{2Ai\varepsilon e^{i\varphi} + B(\varepsilon e^{i2\varphi} - 1)}{\varepsilon e^{i\varphi}(\varepsilon^2 - 1)}$, 留数之和为

$$\begin{aligned} &\frac{-B\varepsilon^2 + B + 2Ai\varepsilon e^{i\varphi} + B(\varepsilon e^{i2\varphi} - 1)}{\varepsilon e^{i\varphi}(\varepsilon^2 - 1)} \\ &= -\frac{2Ai}{1 - \varepsilon^2} - \frac{B\varepsilon(e^{i\varphi} - e^{-i\varphi})}{1 - \varepsilon^2} \\ &= -\frac{2i(A + B\varepsilon \sin \varphi)}{1 - \varepsilon^2} \end{aligned}$$

则

$$I = \frac{1}{2} \cdot 2\pi i \cdot \left[-\frac{2i(A + B\varepsilon \sin \varphi)}{1 - \varepsilon^2} \right] = -\frac{2\pi(A + B\varepsilon \sin \varphi)}{1 - \varepsilon^2}.$$

故

$$\begin{aligned} u(\rho, \varphi) &= \frac{1-(\rho/a)^2}{2\pi} \cdot \frac{2\pi[A + B(\rho/a) \sin \varphi]}{1-(\rho/a)^2} \\ &= A + \frac{B\rho}{a} \sin \varphi. \end{aligned}$$

4. 对于一般的 $f(\varphi)$, 积分公式 (12.2.22) 里的积分可能不那么容易计算. 试将 $1/[a^2 - 2a\rho\cos(\varphi - \varphi_0) + \rho^2]$ 展开为傅里叶级数, 然后逐项积分.

解: 圆内拉氏方程第一边值问题的定解问题是

$$\begin{cases} \Delta_2 u = 0 & (\rho < a), \\ u|_{\rho=a} = f(\varphi). \end{cases} \quad (1)$$

其解的积分公式是

$$\begin{aligned} u(\rho, \varphi) &= \frac{1}{2\pi} \int_0^{2\pi} \frac{a^2 - \rho^2}{a^2 - 2a\rho\cos(\varphi - \varphi_0) + \rho^2} f(\varphi_0) d\varphi_0 \\ &= \frac{1}{2\pi} \int_0^{2\pi} \frac{1 - \varepsilon^2}{1 - 2\varepsilon\cos(\varphi_0 - \varphi) + \varepsilon^2} f(\varphi_0) d\varphi_0 \quad (\varepsilon = \rho/a). \end{aligned} \quad (2)$$

利用 § 5.1 习题 4(2) 的结果

$$\frac{1 - \varepsilon^2}{1 - 2\varepsilon\cos(\varphi_0 - \varphi) + \varepsilon^2} = 1 + 2 \sum_{k=1}^{\infty} \varepsilon^k \cos k(\varphi_0 - \varphi), \quad (3)$$

将③式代入②并逐项积分, 即得

$$\begin{aligned} u(\rho, \varphi) &= \frac{1}{2\pi} \int_0^{2\pi} \left[1 + 2 \sum_{k=1}^{\infty} \varepsilon^k \cos k(\varphi_0 - \varphi) \right] f(\varphi_0) d\varphi_0 \\ &= \sum_{k=1}^{\infty} \frac{1}{\delta_k} \left(\frac{\rho}{a} \right)^k \left[\frac{1}{\pi} \int_0^{2\pi} f(\varphi_0) \cos k\varphi_0 d\varphi_0 \cdot \cos k\varphi \right. \\ &\quad \left. + \frac{1}{\pi} \int_0^{2\pi} f(\varphi_0) \sin k\varphi_0 d\varphi_0 \cdot \sin k\varphi \right] \\ &= \sum_{k=1}^{\infty} (A_k \cos k\varphi + B_k \sin k\varphi) \rho^k \end{aligned}$$

其中

$$\begin{aligned} A_k &= \frac{1}{\delta_k \pi a^k} \int_0^{2\pi} f(\varphi_0) \cos k\varphi_0 d\varphi_0 \\ B_k &= \frac{1}{\pi a^k} \int_0^{2\pi} f(\varphi_0) \sin k\varphi_0 d\varphi_0 \end{aligned}$$

这样, 解是

$$\begin{aligned} u(\rho, \varphi) &= \frac{1}{\pi} \sum_{k=0}^{\infty} \left[\frac{1}{\delta_k} \cos k\varphi \int_0^{2\pi} f(\varphi_0) \cos k\varphi_0 d\varphi_0 \right. \\ &\quad \left. + \sin k\varphi \int_0^{2\pi} f(\varphi_0) \sin k\varphi_0 d\varphi_0 \right] \frac{1}{a^k} \rho^k. \end{aligned}$$

作为对照, 再用分离变数法求解圆的第一边值问题。

$$\begin{cases} \Delta_2 u = 0 & (\rho < a), \\ u|_{\rho=a} = f(\varphi). \end{cases} \quad (4)$$

④的分离变数的一般解是

$$u(\rho, \varphi) = \sum_{k=1}^{\infty} (A_k \cos k\varphi + B_k \sin k\varphi) \rho^k \quad (5)$$

将⑤代入④中边值, 确定系数

$$\sum_{k=1}^{\infty} (A_k a^k \cos k\varphi + B_k a^k \sin k\varphi) = f(\varphi)$$

将 $f(\varphi)$ 展开为傅里叶级数, 比较系数, 有

$$\begin{aligned} A_k &= \frac{1}{\delta_k \pi a^k} \int_0^{2\pi} f(\varphi_0) \cos k\varphi_0 d\varphi_0 \\ B_k &= \frac{1}{\pi a^k} \int_0^{2\pi} f(\varphi_0) \sin k\varphi_0 d\varphi_0 \end{aligned}$$

可见, 两种解法结果相同。

5. 试求层状空间 $0 < z < H$ 第一边值问题的格林函数。

解：格林函数的定解问题是

$$\begin{cases} \Delta G = \delta(x - x_0)\delta(y - y_0)\delta(z - z_0), \\ G|_{z=0} = 0, \quad G|_{z=H} = 0. \end{cases}$$

平面 $z = 0$ 和 $z = H$ 好像两面平面镜，电荷放在其间，被平面镜反复反射，造成无限多电像（图 12-3）。对于 (x_0, y_0, z_0) 处的点电荷，所有 $(x_0, y_0, 2nH + z_0)$ 处的电像与其带同号电荷，所有 $(x_0, y_0, 2nH - z_0)$ 处的电像与其带异号电荷。这样，利用半空间的格林函数（参见 § 12.2 例 2），并对各层叠加，结果是

$$G(r, r_0) = -\frac{1}{4\pi} \sum_{n=-\infty}^{\infty} \frac{1}{\sqrt{(x-x_0)^2 + (y-y_0)^2 + (z-2nH-z_0)^2}} \\ + \frac{1}{4\pi} \sum_{n=-\infty}^{\infty} \frac{1}{\sqrt{(x-x_0)^2 + (y-y_0)^2 + (z-2nH+z_0)^2}}.$$

虽然积分公式的形式是有限的，格林函数本身却是无穷级数。

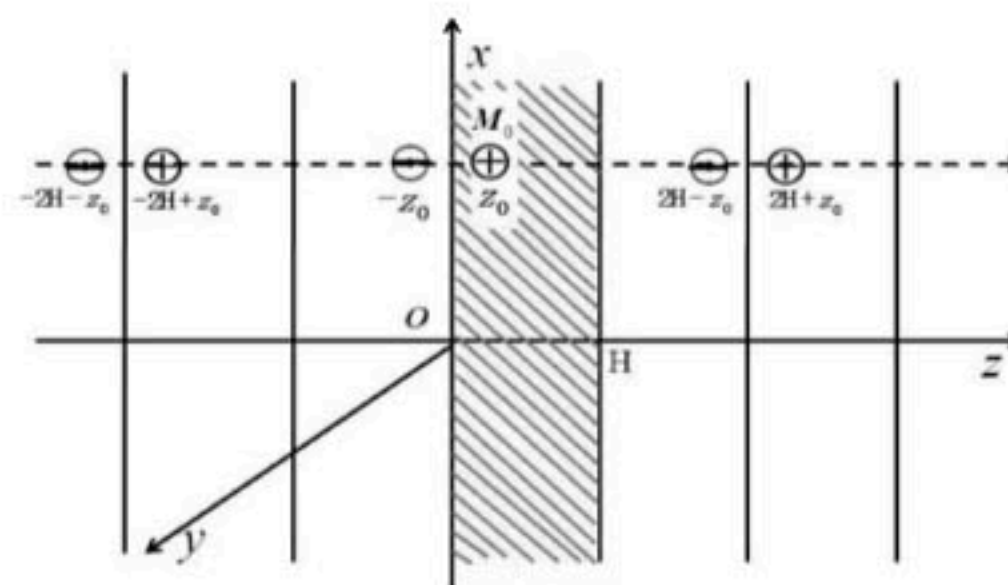


图12-3

§ 12.4 用冲量定理法求格林函数

1. 两端固定的弦在线密度为 $\rho f(x, t) = \rho \Phi(x) \sin \omega t$ 的横向力作用下振动. 求解其振动情况. 研究共振的可能性, 并求共振时的解.

解: 定解问题为

$$\begin{cases} u_{tt} - a^2 u_{xx} = \Phi(x) \sin \omega t, \\ u|_{x=0} = 0, u|_{x=l} = 0, \\ u|_{t=0} = 0, u_t|_{t=0} = 0. \end{cases} \quad (1)$$

①的格林函数 G 满足

$$\begin{cases} G_{tt} - a^2 G_{xx} = \delta(x - \xi) \delta(t - \tau), \\ G|_{x=0} = 0, G|_{x=l} = 0, \\ G|_{t=0} = 0, G_t|_{t=0} = 0. \end{cases} \quad (2)$$

按冲量定理, ②可转化为

$$\begin{cases} G_{tt} - a^2 G_{xx} = 0, \\ G|_{x=0} = 0, G|_{x=l} = 0, \\ G|_{t=\tau+0} = 0, G_t|_{t=\tau+0} = \delta(x - \xi). \end{cases} \quad (3)$$

利用分离变数法, ③的一般解为

$$G(x, t; \xi, \tau) = \sum_{n=1}^{\infty} \left[A_n(\xi, \tau) \cos \frac{n\pi a(t - \tau)}{l} + B_n(\xi, \tau) \sin \frac{n\pi a(t - \tau)}{l} \right] \sin \frac{n\pi x}{l}. \quad (4)$$

将④代入③中初始条件, 有

$$\begin{cases} A_n(\xi, \tau) = 0, \\ B_n(\xi, \tau) = \frac{2}{n\pi a} \int_0^l \delta(x - \xi) \sin \frac{n\pi x}{l} dx = \frac{2}{n\pi a} \sin \frac{n\pi \xi}{l}. \end{cases}$$

这样

$$G(x, t; \xi, \tau) = \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi a(t - \tau)}{l} \sin \frac{n\pi \xi}{l} \sin \frac{n\pi x}{l}. \quad (5)$$

将⑤代入 (12.3.19), 得

$$\begin{aligned} u(x, t) &= \int_{\tau=0}^t \int_{\xi=0}^l f(\xi, \tau) G(x, t; \xi, \tau) d\xi d\tau \\ &= \int_{\tau=0}^t \int_{\xi=0}^l \left[\Phi(\xi) \sin \omega \tau \cdot \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi a(t - \tau)}{l} \sin \frac{n\pi \xi}{l} \sin \frac{n\pi x}{l} \right] d\xi d\tau \\ &= \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi x}{l} \int_{\xi=0}^l \Phi(\xi) \sin \frac{n\pi \xi}{l} d\xi \int_{\tau=0}^t \sin \omega \tau \sin \frac{n\pi a(t - \tau)}{l} d\tau \\ &= \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \int_0^l \Phi(\xi) \sin \frac{n\pi \xi}{l} d\xi \left(\frac{\omega \sin \frac{n\pi a}{l} t - \frac{n\pi a}{l} \sin \omega t}{\omega^2 - n^2 \pi^2 a^2 / l^2} \right) \sin \frac{n\pi x}{l}, \end{aligned} \quad (6)$$

如外力的频率等于基音或谐音的频率, 则⑥中[]内为 $0/0$ 型, 按洛必达法则其极限为 $\left(\frac{1}{2\omega} \sin \omega t - \frac{1}{2} t \cos \omega t \right)$, 其中第二部分的振幅为 $\frac{1}{2} t$, 随时间而增长, 这就是共振。

2. 两端固定的弦在点 x_0 受谐变力 $\rho f(t) = \rho f_0 \sin \omega t$ 作用而振动, 求解振动情况。

解: 定解问题

$$\begin{cases} u_{tt} - a^2 u_{xx} = f_0 \sin \omega t \delta(x - x_0), \\ u|_{x=0} = 0, u|_{x=l} = 0, \\ u|_{t=0} = 0, u_t|_{t=0} = 0. \end{cases} \quad (1)$$

①的格林函数 G 满足

$$\begin{cases} G_{tt} - a^2 G_{xx} = \delta(x - \xi) \delta(t - \tau), \\ G|_{x=0} = 0, G|_{x=l} = 0, \\ G|_{t=0} = 0, G_t|_{t=0} = 0. \end{cases} \quad (2)$$

按冲量定理, ②可转化为

$$\begin{cases} G_{tt} - a^2 G_{xx} = 0, \\ G|_{x=0} = 0, G|_{x=l} = 0, \\ G|_{t=\tau+0} = 0, G_t|_{t=\tau+0} = \delta(x - \xi). \end{cases} \quad (3)$$

利用分离变数法, ③的一般解为

$$G(x, t; \xi, \tau) = \sum_{n=1}^{\infty} \left[A_n(\xi, \tau) \cos \frac{n\pi a(t - \tau)}{l} + B_n(\xi, \tau) \sin \frac{n\pi a(t - \tau)}{l} \right] \sin \frac{n\pi x}{l}. \quad (4)$$

将④代入③中初始条件, 有

$$\begin{cases} A_n(\xi, \tau) = 0, \\ B_n(\xi, \tau) = \frac{2}{n\pi a} \int_0^l \delta(x - \xi) \sin \frac{n\pi x}{l} dx = \frac{2}{n\pi a} \sin \frac{n\pi \xi}{l}. \end{cases}$$

这样

$$G(x, t; \xi, \tau) = \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi a(t - \tau)}{l} \sin \frac{n\pi \xi}{l} \sin \frac{n\pi x}{l}. \quad (5)$$

将⑤代入 (12.3.19), 得

$$\begin{aligned} u(x, t) &= \int_{\tau=0}^t \int_{\xi=0}^l f(\xi, \tau) G(x, t; \xi, \tau) d\xi d\tau \\ &= \int_{\tau=0}^t \int_{\xi=0}^l \left[f_0 \delta(\xi - x_0) \sin \omega \tau \cdot \frac{2}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi a(t - \tau)}{l} \sin \frac{n\pi \xi}{l} \sin \frac{n\pi x}{l} \right] d\xi d\tau \\ &= \frac{2f_0}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi x}{l} \int_{\xi=0}^l \delta(\xi - x_0) \sin \frac{n\pi \xi}{l} d\xi \int_{\tau=0}^t \sin \omega \tau \sin \frac{n\pi a(t - \tau)}{l} d\tau \\ &= \frac{2f_0}{\pi a} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi x_0}{l} \left(\frac{\omega \sin \frac{n\pi a}{l} t - \frac{n\pi a}{l} \sin \omega t}{\omega^2 - n^2 \pi^2 a^2 / l^2} \right) \sin \frac{n\pi x}{l}. \end{aligned}$$

3. 长为 l 的均匀细导线, 每单位长的电阻为 R , 通过恒定的电流 I , 导线表面跟周围温度为零度的介质进行热量交换. 试求解导线上的温度变化. 设初始温度和导线两端温度都是零度.

解: 泛定方程为 $c\rho u_t - ku_{xx} = -hu + I^2 R$, 即

$$u_t - a^2 u_{xx} + \frac{h}{c\rho} u = \frac{1}{c\rho} I^2 R.$$

从而, 定解问题为

$$\begin{cases} u_t - a^2 u_{xx} + \frac{h}{c\rho} u = \frac{1}{c\rho} I^2 R, \\ u|_{x=0} = 0, \quad u|_{x=l} = 0, \\ u|_{t=0} = 0. \end{cases} \quad (1)$$

先求格林函数 G , G 满足

$$\begin{cases} G_t - a^2 G_{xx} + \frac{h}{c\rho} G = \delta(x - \xi)\delta(t - \tau), \\ G|_{x=0} = 0, \quad G|_{x=l} = 0, \\ G|_{t=0} = 0. \end{cases} \quad (2)$$

按冲量定理, 这个问题可转化为

$$\begin{cases} G_t - a^2 G_{xx} + \frac{h}{c\rho} G = 0, \\ G|_{x=0} = 0, \quad G|_{x=l} = 0, \\ G|_{t=\tau+0} = \delta(x - \xi). \end{cases} \quad (3)$$

利用分离变数法求解③, 有

$$G(x, t; \xi, \tau) = \sum_{n=1}^{\infty} A_n(\xi, \tau) \exp \left[- \left(\frac{n^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right) (t - \tau) \right] \sin \frac{n\pi x}{l}. \quad (4)$$

将④代入③中初始条件, 有

$$\sum_{n=1}^{\infty} A_n(\xi, \tau) \sin \frac{n\pi x}{l} = \delta(x - \xi),$$

其中 $A_n(\xi, \tau) = \frac{1}{l} \sin \frac{n\pi \xi}{l}$. 于是

$$G(x, t; \xi, \tau) = \sum_{n=1}^{\infty} \frac{2}{l} \exp \left[- \left(\frac{n^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right) (t - \tau) \right] \sin \frac{n\pi \xi}{l} \sin \frac{n\pi x}{l}.$$

解是

$$\begin{aligned} u(x, t) &= \int_{\tau=0}^t \int_{\xi=0}^l f(\xi, \tau) G(x, t; \xi, \tau) d\xi d\tau \\ &= \int_{\tau=0}^t \int_{\xi=0}^l \left\{ \frac{1}{c\rho} I^2 R \cdot \sum_{n=1}^{\infty} \frac{2}{l} \exp \left[- \left(\frac{n^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right) (t - \tau) \right] \sin \frac{n\pi \xi}{l} \sin \frac{n\pi x}{l} \right\} d\xi d\tau \\ &= \frac{2I^2 R}{lc\rho} \sum_{n=1}^{\infty} \int_{\xi=0}^l \sin \frac{n\pi x}{l} \sin \frac{n\pi \xi}{l} d\xi \int_{\tau=0}^t \exp \left[- \left(\frac{n^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right) (t - \tau) \right] d\tau \\ &= \frac{2I^2 R}{lc\rho} \sum_{n=1}^{\infty} \frac{l}{n\pi} [1 - (-1)^n] \frac{1}{\frac{n^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho}} \left\{ 1 - \exp \left[- \left(\frac{n^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right) t \right] \right\} \sin \frac{n\pi x}{l} \end{aligned}$$

n 为奇数时解才不为零. 将 n 换成 $2n+1$, 得解为

$$u(x, t) = \frac{4I^2 R}{\pi c\rho} \sum_{n=0}^{\infty} \frac{1}{2n+1} \frac{1}{(2n+1)^2 \pi^2 a^2 / l^2 + h / c\rho} \sin \frac{(2n+1)\pi x}{l} \left[1 - e^{-\left(\frac{(2n+1)^2 \pi^2 a^2}{l^2} + \frac{h}{c\rho} \right) t} \right].$$

说明: 以下第 4 至第 6 题求解中出现误差函数 $\operatorname{erf}(x)$ 及余误差函数 $\operatorname{erfc}(x)$. 关于这两个函数, 具体请参见 § 13.1 例 5.

4. 求解一维半无界空间的输运问题

$$\begin{cases} u_t - a^2 u_{xx} = 0, \\ u|_{x=0} = At, \\ u|_{t=0} = 0. \end{cases} \quad (1)$$

解: 令 $u = w + At$, 定解问题①转化成 w 的定解问题

$$\begin{cases} w_t - a^2 w_{xx} = -A, \\ w|_{x=0} = 0, \\ w|_{t=0} = 0. \end{cases} \quad (2)$$

先将上述定解问题延拓到 $x < 0$ 的半空间

$$\begin{cases} w_t - a^2 w_{xx} = \begin{cases} -A & x > 0 \\ A & x < 0 \end{cases}, \\ w|_{t=0} = 0. \end{cases} \quad (3)$$

先求上式的格林函数, 其满足定解问题

$$\begin{cases} G_t - a^2 G_{xx} = \delta(x - \xi)\delta(t - \tau), \\ G|_{t=0} = 0. \end{cases}$$

由冲量定理, 这个问题又可转化成

$$\begin{cases} G_t - a^2 G_{xx} = 0, \\ G|_{t=\tau} = \delta(x - \xi). \end{cases}$$

这个定解问题的解可引用 § 13.1 例 2 的结果, 只是那里的 t 在这里应换为 $t - \tau$. 于是, 得到无界空间输运问题的格林函数

$$\begin{aligned} G(x, t; \xi, \tau) &= \int_{-\infty}^{\infty} \delta(x_0 - \xi) \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-x_0)^2}{4a^2(t-\tau)}} \right] dx_0 \\ &= \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}}. \end{aligned}$$

从而 w 的解

$$\begin{aligned} w(x, t) &= \int_{\tau=0}^t \int_{\xi=-\infty}^0 A \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} \right] d\xi d\tau \\ &\quad + \int_{\tau=0}^t \int_{\xi=0}^{\infty} (-A) \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} \right] d\xi d\tau. \end{aligned}$$

在上式右边第一个积分中, 作变换 $z = \frac{\xi - x}{2a\sqrt{t-\tau}}$, 第二个积分中, 作变换 $z = \frac{x - \xi}{2a\sqrt{t-\tau}}$, 有

$$\begin{aligned} w(x, t) &= -\int_{\tau=0}^t \left[\frac{A}{\sqrt{\pi}} \int_{\frac{x}{2a\sqrt{t-\tau}}}^{\infty} e^{-z^2} dz + \frac{A}{\sqrt{\pi}} \int_{-\infty}^{\frac{-x}{2a\sqrt{t-\tau}}} e^{-z^2} dz \right] d\tau \\ &= -A \int_{\tau=0}^t \left[\frac{2}{\sqrt{\pi}} \int_0^{\frac{x}{2a\sqrt{t-\tau}}} e^{-z^2} dz \right] d\tau \\ &= -A \int_{\tau=0}^t \operatorname{erf} \left(\frac{x}{2a\sqrt{t-\tau}} \right) d\tau \end{aligned}$$

问题的解是

$$\begin{aligned} u(x, t) &= At - A \int_0^t \operatorname{erf} \left(\frac{x}{2a\sqrt{t-\tau}} \right) d\tau \\ &= A \int_0^t \left[1 - \operatorname{erf} \left(\frac{x}{2a\sqrt{t-\tau}} \right) \right] d\tau \\ &= A \int_0^t \operatorname{erfc} \left(\frac{x}{2a\sqrt{t-\tau}} \right) d\tau. \end{aligned}$$

5. 在一维半无界空间中求解

$$\begin{cases} u_t - a^2 u_{xx} = 0, \\ u|_{x=0} = f(t), \\ u|_{t=0} = \varphi(x). \end{cases} \quad (1)$$

解: 令 $u = w + f(t)$, 定解问题①转化成 w 的定解问题

$$\begin{cases} w_t - a^2 w_{xx} = -f'(t), \\ w|_{x=0} = 0, \\ w|_{t=0} = \varphi(x) - f(0). \end{cases} \quad (2)$$

先将定解问题②延拓到 $x < 0$ 的半空间

$$\begin{cases} w_t - a^2 w_{xx} = \begin{cases} -f'(t), & (x > 0) \\ f'(t), & (x < 0) \end{cases} \\ w|_{t=0} = \begin{cases} \varphi(x) - f(0), & (x > 0) \\ -\varphi(-x) + f(0), & (x < 0) \end{cases} \end{cases} \quad (3)$$

先求③的格林函数 G , 其满足定解问题

$$\begin{cases} G_t - a^2 G_{xx} = \delta(x - \xi)\delta(t - \tau), \\ G|_{t=0} = 0. \end{cases} \quad (4)$$

由冲量定理, ④又可转化成

$$\begin{cases} G_t - a^2 G_{xx} = 0, \\ G|_{t=\tau} = \delta(x - \xi). \end{cases}$$

这个定解问题的解可引用 § 13.1 例 2 的结果, 只是那里的 t 在这里应换为 $t - \tau$ 。于是, 得到无界空间输运问题的格林函数

$$\begin{aligned} G(x, t; \xi, \tau) &= \int_{-\infty}^{\infty} \delta(x_0 - \xi) \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-x_0)^2}{4a^2(t-\tau)}} \right] dx_0 \\ &= \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-(x-\xi)^2/4a^2(t-\tau)}. \end{aligned}$$

接着求 w 的解, 利用公式 (12.3.32)

$$w(x, t) = \int_{-\infty}^{\infty} \int_0^t G(x, t; \xi, \tau) f'(\xi, \tau) d\xi d\tau + \int_{-\infty}^{\infty} (Gw)_{\tau=0} d\xi.$$

其中

$$(Gw)_{\tau=0} = \frac{1}{2a\sqrt{\pi t}} e^{-(x-\xi)^2/4a^2 t} \begin{cases} \varphi(x) - f(0) & (x > 0) \\ -\varphi(-x) + f(0) & (x < 0) \end{cases}$$

这样

$$\begin{aligned}
w(x,t) = & \int_{\tau=0}^t \int_{\xi=-\infty}^0 f'(\tau) \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} \right] d\xi d\tau \\
& + \int_{\tau=0}^t \int_{\xi=0}^{\infty} [-f'(\tau)] \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} \right] d\xi d\tau \\
& + \int_{-\infty}^0 [-\varphi(-x) + f(0)] \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi \\
& + \int_0^{\infty} [\varphi(x) - f(0)] \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi. \quad (5)
\end{aligned}$$

在⑤式右边第一个积分中, 作变换 $z = \frac{x-\xi}{2a\sqrt{t-\tau}}$, 则第一个积分可化为

$$-\int_{\tau=0}^t \int_{\xi=-\infty}^0 f'(\tau) \frac{1}{\sqrt{\pi}} e^{-z^2} dz d\tau$$

在⑤式右边第二个积分中, 作变换 $z = \frac{\xi-x}{2a\sqrt{t-\tau}}$, 则第二个积分可化为

$$-\int_{\tau=0}^t \int_{\xi=0}^{\infty} f'(\tau) \frac{1}{\sqrt{\pi}} e^{-z^2} dz d\tau$$

这样, 前两项积分合并为

$$-\int_{\tau=0}^t f'(\tau) \int_{\frac{-x}{2a\sqrt{t-\tau}}}^{\frac{x}{2a\sqrt{t-\tau}}} \frac{1}{\sqrt{\pi}} e^{-z^2} dz d\tau = -\int_{\tau=0}^t f'(\tau) \operatorname{erf} \left(\frac{x}{2a\sqrt{t-\tau}} \right) d\tau$$

在⑤式右边第三项积分中, 作变换 $z = \frac{x-\xi}{2a\sqrt{t}}$, 则第三个积分可化为

$$-\int_{-\infty}^0 \frac{1}{2a\sqrt{t}} [-\varphi(-x+2a\sqrt{t}z) + f(0)] \frac{1}{\sqrt{\pi}} e^{-z^2} dz.$$

在⑤式右边第四项积分中, 作变换 $z = \frac{\xi-x}{2a\sqrt{t}}$, 则第四个积分可化为

$$-\int_{\frac{-x}{2a\sqrt{t}}}^{\infty} [\varphi(x+2a\sqrt{t}z) - f(0)] \frac{1}{\sqrt{\pi}} e^{-z^2} dz.$$

这样, 后两项积分合并为

$$\begin{aligned}
& -f(0) \left(\int_{\frac{x}{2a\sqrt{t}}}^{\infty} \frac{1}{\sqrt{\pi}} e^{-z^2} dz + \int_{\frac{-x}{2a\sqrt{t}}}^{\infty} \frac{1}{\sqrt{\pi}} e^{-z^2} dz \right) \\
& + \int_{\frac{-x}{2a\sqrt{t}}}^{\infty} [\varphi(x+2a\sqrt{t}z)] \frac{1}{\sqrt{\pi}} e^{-z^2} dz \\
& - \int_{\frac{x}{2a\sqrt{t}}}^{\infty} [-\varphi(-x+2a\sqrt{t}z)] \frac{1}{\sqrt{\pi}} e^{-z^2} dz
\end{aligned}$$

进一步合并, 得

$$-f(0)\operatorname{erf}\left(\frac{x}{2a\sqrt{t}}\right)+\int_{\frac{-x}{2a\sqrt{t}}}^{\infty}\left[\varphi(x+2a\sqrt{t}z)\right]\frac{1}{\sqrt{\pi}}e^{-z^2}dz$$

$$-\int_{\frac{x}{2a\sqrt{t}}}^{\infty}\left[-\varphi(-x+2a\sqrt{t}z)\right]\frac{1}{\sqrt{\pi}}e^{-z^2}dz$$

这样,

$$w(x,t)=-\int_{\tau=0}^tf'(\tau)\operatorname{erf}\left(\frac{-x}{2a\sqrt{t-\tau}}\right)d\tau-f(0)\operatorname{erf}\left(\frac{x}{2a\sqrt{t}}\right)$$

$$+\int_{\frac{-x}{2a\sqrt{t}}}^{\infty}\left[\varphi(x+2a\sqrt{t}z)\right]\frac{1}{\sqrt{\pi}}e^{-z^2}dz$$

$$-\int_{\frac{x}{2a\sqrt{t}}}^{\infty}\left[\varphi(-x+2a\sqrt{t}z)\right]\frac{1}{\sqrt{\pi}}e^{-z^2}dz$$

结果

$$u(x,t)=f(t)-\int_{\tau=0}^tf'(\tau)\operatorname{erf}\left(\frac{-x}{2a\sqrt{t-\tau}}\right)d\tau-f(0)\operatorname{erf}\left(\frac{x}{2a\sqrt{t}}\right)$$

$$+\int_{\frac{-x}{2a\sqrt{t}}}^{\infty}\left[\varphi(x+2a\sqrt{t}z)\right]\frac{1}{\sqrt{\pi}}e^{-z^2}dz$$

$$-\int_{\frac{x}{2a\sqrt{t}}}^{\infty}\left[-\varphi(-x+2a\sqrt{t}z)\right]\frac{1}{\sqrt{\pi}}e^{-z^2}dz$$

其中 $f(t)$ 可写成 $-\int_0^t f'(\tau)d\tau + f(0)$, 则

$$f(t)-\int_{\tau=0}^tf'(\tau)\operatorname{erf}\left(\frac{-x}{2a\sqrt{t-\tau}}\right)d\tau-f(0)\operatorname{erf}\left(\frac{x}{2a\sqrt{t}}\right)$$

$$=\int_{\tau=0}^tf'(\tau)\left[1-\operatorname{erf}\left(\frac{-x}{2a\sqrt{t-\tau}}\right)\right]d\tau-f(0)\left[1-\operatorname{erf}\left(\frac{x}{2a\sqrt{t}}\right)\right]$$

$$=\int_{\tau=0}^tf'(\tau)\operatorname{erfc}\left(\frac{-x}{2a\sqrt{t-\tau}}\right)d\tau-f(0)\operatorname{erfc}\left(\frac{x}{2a\sqrt{t}}\right)$$

于是

$$u(x,t)=f(0)\operatorname{erfc}\left(\frac{x}{2a\sqrt{t}}\right)+\int_0^tf'(\tau)\operatorname{erfc}\left(\frac{x}{2a\sqrt{t-\tau}}\right)d\tau$$

$$+\int_{-x/2a\sqrt{t}}^{\infty}\varphi(x+2az\sqrt{t})\frac{1}{\sqrt{\pi}}e^{-z^2}dz-\int_{x/2a\sqrt{t}}^{\infty}\varphi(2az\sqrt{t}-x)\frac{1}{\sqrt{\pi}}e^{-z^2}dz。$$

6. 在一维半无界空间中求解

$$\begin{cases} u_t - a^2 u_{xx} = 0, \\ u_x|_{x=0} = q(t), \\ u|_{t=0} = 0. \end{cases} \quad (1)$$

解: 令 $u = w - e^{-x}q(t)$, 定解问题①转化成 w 的定解问题

$$\begin{cases} w_t - a^2 w_{xx} = q'(t)e^{-x} - a^2 q(t)e^{-x}, \\ w_x|_{x=0} = 0, \\ w|_{t=0} = q(0)e^{-x}. \end{cases} \quad (2)$$

根据第二类齐次边界条件, 先将问题②偶延拓到 $x < 0$ 的半空间

$$\begin{cases} w_t - a^2 w_{xx} = \begin{cases} [q'(t) - a^2 q(t)]e^{-x}, & (x > 0) \\ [q'(t) - a^2 q(t)]e^x, & (x < 0) \end{cases} \\ w|_{t=0} = \begin{cases} q(0)e^{-x}, & (x > 0) \\ q(0)e^x, & (x < 0) \end{cases} \end{cases} \quad (3)$$

先求③的格林函数 G , 其满足定解问题

$$\begin{cases} G_t - a^2 G_{xx} = \delta(x - \xi)\delta(t - \tau), \\ G|_{t=0} = 0. \end{cases} \quad (4)$$

由冲量定理, 这个问题又可转化成

$$\begin{cases} G_t - a^2 G_{xx} = 0, \\ G|_{t=\tau} = \delta(x - \xi). \end{cases}$$

这个定解问题的解可引用 § 13.1 例 2 的结果, 只是那里的 t 在这里应换为 $t - \tau$ 。于是, 得到无界空间输运问题的格林函数

$$\begin{aligned} G(x, t; \xi, \tau) &= \int_{-\infty}^{\infty} \delta(x_0 - \xi) \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-x_0)^2}{4a^2(t-\tau)}} \right] dx_0 \\ &= \frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-(x-\xi)^2/4a^2(t-\tau)}. \end{aligned}$$

接着求 w 的解, 利用公式 (12.3.32)

$$w(x, t) = \int_{-\infty}^{\infty} \int_0^t G(x, t; \xi, \tau) f(\xi, \tau) d\xi d\tau + \int_{-\infty}^{\infty} (Gw)_{\tau=0} d\xi.$$

$$\text{其中 } (Gw)_{\tau=0} = \frac{1}{2a\sqrt{\pi t}} e^{-(x-\xi)^2/4a^2 t} \begin{cases} q(0)e^{-x} & (x > 0) \\ q(0)e^x & (x < 0) \end{cases}$$

于是

$$\begin{aligned}
w(x,t) = & \int_{\tau=0}^t \int_{\xi=-\infty}^0 [q'(\tau) - a^2 q(\tau)] e^{\xi} \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} \right] d\xi d\tau \\
& + \int_{\tau=0}^t \int_{\xi=0}^{\infty} [q'(\tau) - a^2 q(\tau)] e^{-\xi} \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{(x-\xi)^2}{4a^2(t-\tau)}} \right] d\xi d\tau \\
& + \int_{-\infty}^0 q(0) e^{\xi} \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi \\
& + \int_0^{\infty} q(0) e^{-\xi} \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4a^2 t}} d\xi, \tag{5}
\end{aligned}$$

对前两项积分, 对 e 指数部分进行配平方处理, 而改写成

$$\begin{aligned}
& \int_{\tau=0}^t \int_{\xi=-\infty}^0 [q'(\tau) - a^2 q(\tau)] \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{\{[x+2a^2(t-\tau)]-\xi\}^2}{4a^2(t-\tau)}} \right] e^{a^2(t-\tau)} e^x d\xi d\tau \\
& + \int_{\tau=0}^t \int_{\xi=0}^{\infty} [q'(\tau) - a^2 q(\tau)] \left[\frac{1}{2a\sqrt{\pi(t-\tau)}} e^{-\frac{\{[x-2a^2(t-\tau)]-\xi\}^2}{4a^2(t-\tau)}} \right] e^{a^2(t-\tau)} e^{-x} d\xi d\tau
\end{aligned}$$

在第一项积分中令 $z = \frac{[x+2a^2(t-\tau)]-\xi}{2a\sqrt{t-\tau}}$, 则第一项积分可化为

$$\begin{aligned}
& \int_{\tau=0}^t \int_{-\infty}^{\frac{x+2a^2(t-\tau)}{2a\sqrt{t-\tau}}} [q'(\tau) - a^2 q(\tau)] \left[\frac{1}{\sqrt{\pi}} e^{-z^2} \right] e^{a^2(t-\tau)} e^x dz d\tau \\
& = \int_{\tau=0}^t -\frac{1}{2} [q'(\tau) - a^2 q(\tau)] e^{a^2(t-\tau)} e^x \left[\int_0^{\frac{x+2a^2(t-\tau)}{2a\sqrt{t-\tau}}} \frac{2}{\sqrt{\pi}} e^{-z^2} dz + \int_{\infty}^0 \frac{2}{\sqrt{\pi}} e^{-z^2} dz \right] d\tau \\
& = \int_{\tau=0}^t -\frac{1}{2} [q'(\tau) - a^2 q(\tau)] e^{a^2(t-\tau)} e^x \left\{ \operatorname{erf} \left[\frac{x+2a^2(t-\tau)}{2a\sqrt{t-\tau}} \right] - 1 \right\} d\tau \\
& = \int_{\tau=0}^t \frac{1}{2} [q'(\tau) - a^2 q(\tau)] e^{a^2(t-\tau)} e^x \operatorname{erfc} \left[\frac{x+2a^2(t-\tau)}{2a\sqrt{t-\tau}} \right] d\tau
\end{aligned}$$

在第二项积分中令 $z = \frac{\xi - [x-2a^2(t-\tau)]}{2a\sqrt{t-\tau}}$, 则第二项积分可化为

$$\begin{aligned}
& \int_{\tau=0}^t \int_{\frac{x-2a^2(t-\tau)}{2a\sqrt{t-\tau}}}^0 -[q'(\tau)-a^2q(\tau)] \left(\frac{1}{\sqrt{\pi}} e^{-z^2} \right) e^{a^2(t-\tau)} e^{-x} d\xi d\tau \\
&= \int_{\tau=0}^t \frac{1}{2} [q'(\tau)-a^2q(\tau)] e^{a^2(t-\tau)} e^{-x} \left[\int_{\frac{x-2a^2(t-\tau)}{2a\sqrt{t-\tau}}}^0 \frac{2}{\sqrt{\pi}} e^{-z^2} dz + \int_0^{\infty} \frac{2}{\sqrt{\pi}} e^{-z^2} dz \right] d\tau \\
&= \int_{\tau=0}^t \frac{1}{2} [q'(\tau)-a^2q(\tau)] e^{a^2(t-\tau)} e^{-x} \left\{ -\operatorname{erf} \left[\frac{x-2a^2(t-\tau)}{2a\sqrt{t-\tau}} \right] + 1 \right\} d\tau \\
&= \int_{\tau=0}^t \frac{1}{2} [q'(\tau)-a^2q(\tau)] e^{a^2(t-\tau)} e^{-x} \operatorname{erfc} \left[\frac{x-2a^2(t-\tau)}{2a\sqrt{t-\tau}} \right] d\tau
\end{aligned}$$

前两项积分可合并为

$$\int_{\tau=0}^t \frac{1}{2} [q'(\tau)-a^2q(\tau)] e^{a^2(t-\tau)} \left\{ e^x \operatorname{erfc} \left[\frac{x+2a^2(t-\tau)}{2a\sqrt{t-\tau}} \right] + e^{-x} \operatorname{erfc} \left[\frac{x-2a^2(t-\tau)}{2a\sqrt{t-\tau}} \right] \right\} d\tau$$

在第三、第四项积分中, 分别令 $z = \frac{(x+2a^2t)-\xi}{2a\sqrt{t}}$, 及 $z = \frac{\xi-(x-2a^2t)}{2a\sqrt{t}}$, 并进行与第一、二项积分中类似的变换, 可将第三第四项积分合并为

$$\frac{1}{2} q(0) e^{a^2 t} \left[e^x \operatorname{erfc} \left(\frac{2a^2 t + x}{2a\sqrt{t}} \right) + e^{-x} \operatorname{erfc} \left(\frac{2a^2 t - x}{2a\sqrt{t}} \right) \right]$$

最终结果为

$$\begin{aligned}
u(x,t) &= -e^{-x} q(t) + w(x,t) \\
&= -e^{-x} q(t) + \frac{1}{2} q(0) e^{a^2 t} \left\{ e^x \operatorname{erfc} \left(\frac{2a^2 t + x}{2a\sqrt{t}} \right) \right. \\
&\quad \left. + e^{-x} \operatorname{erfc} \left(\frac{2a^2 t - x}{2a\sqrt{t}} \right) \right\} + \int_0^t \frac{1}{2} [q'(\tau)-a^2q(\tau)] e^{a^2(t-\tau)} \\
&\quad \times \left\{ e^x \operatorname{erfc} \left(\frac{2a^2(t-\tau)+x}{2a\sqrt{t-\tau}} \right) + e^{-x} \operatorname{erfc} \left(\frac{2a^2(t-\tau)-x}{2a\sqrt{t-\tau}} \right) \right\} d\tau. \quad (x > 0)
\end{aligned}$$

* § 12.5 推广的格林公式及其应用

1. 求解 $u_{tt} - a^2 u_{xx} = f(x, t), u|_{t=0} = \varphi(x), u_t|_{t=0} = \psi(x)$ 。

解：首先，将原问题重写成如下定解问题

$$\begin{cases} u_{tt} - a^2 u_{xx} = f(x, t), & \text{①} \\ u|_{t=0} = \varphi(x), & \text{②} \\ u_t|_{t=0} = \psi(x). & \text{③} \end{cases}$$

根据教材 (7.3.12) (7.3.13) 判断, $a_{12}^2 - a_{11}a_{22} = a^2 > 0$, 故本题方程①为含有两个自变数的常系数双曲型方程。首先, 将方程①化成形为

$$Lu \equiv u_{xx} - u_{yy} + b_1 u_x + b_2 u_y + cu = f$$

的标准型。①的特征线是 $x \pm at = \text{常数}$ 。作变数变换

$$\begin{cases} x = x \\ t = \frac{y}{a} \end{cases}$$

则原定解问题变换成

$$\begin{cases} u_{xx} - u_{yy} = -\frac{1}{a^2} f(x, \frac{y}{a}), & \text{④} \\ u|_{y=0} = \varphi(x), & \text{⑤} \\ u_y|_{y=0} = \frac{1}{a} \psi(x). & \text{⑥} \end{cases}$$

本题中, $a_{11} = 1, a_{22} = -1, a_{12} = b_1 = b_2 = c = 0, L = M = \frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2}$, 是自伴的, 黎曼函数 v 是 $Mv = 0$ 的解, 而且满足

$$\begin{cases} \frac{\partial v}{\partial s} + \frac{b_2 + b_1}{2\sqrt{2}} v = 0 & (\text{在 } M_2 M_0 \text{ 上}) \\ \frac{\partial v}{\partial s} - \frac{b_2 - b_1}{2\sqrt{2}} v = 0 & (\text{在 } M_0 M_1 \text{ 上}) \\ v(x_0, y_0) = 1 \end{cases}$$

$$\text{从而有 } \begin{cases} v = e^{\int_{s_0}^s \frac{b_2 + b_1}{2\sqrt{2}} ds} = e^0 = 1 & (\text{在 } M_2 M_0 \text{ 上}), \\ v = e^{\int_{s_0}^s \frac{b_2 - b_1}{2\sqrt{2}} ds} = e^0 = 1 & (\text{在 } M_0 M_1 \text{ 上}). \end{cases}$$

这样, 利用推广的格林公式, 本问题的一般解是

$$\begin{aligned}
u(x_0, y_0) &= \frac{1}{2} [u(x_1, y_1)v(x_1, y_1) + u(x_2, y_2)v(x_2, y_2)] \\
&\quad + \frac{1}{2} \int_{M_1 M_2} [X \cos(\mathbf{n}, \mathbf{x}) + Y \cos(\mathbf{n}, \mathbf{y})] ds \\
&\quad - \frac{1}{2} \iint_{M_0 M_1 M_2} v(x, y, z; x_0, y_0, z_0) f(x, y, z) ds.
\end{aligned} \tag{7}$$

下面分别计算出⑦中各项 (参见图 12-4),

$$u(x_1, y_1) = u(x_1, 0) = \varphi(x_0 - at_0),$$

$$u(x_2, y_2) = u(x_2, 0) = \varphi(x_0 + at_0),$$

$$v(x_1, y_1) = v(x_2, y_2) = 1,$$

所以, ⑦式右端第一项为

$$\frac{1}{2} [\varphi(x_0 - at_0) + \varphi(x_0 + at_0)].$$

由教材 (12.5.30) 式

$$\begin{cases} X = -(uv)_x + (2u_x + b_1 u)v = u_x, \\ Y = (uv)_y - (2u_y - b_2 u)v = -u_y. \end{cases}$$

$$\cos(\mathbf{n}, \mathbf{x}) = 0, \quad \cos(\mathbf{n}, \mathbf{y}) = -1,$$

所以, ⑦式右端第二项为

$$\frac{1}{2} \int_{M_1}^{M_2} u_y|_{y=0} dx = \frac{1}{2a} \int_{x_0 - at_0}^{x_0 + at_0} \psi(\xi) d\xi.$$

在区域 $M_0 M_1 M_2$ 内, $v=1$, $dy=adt$, ⑦式右端第三项为

$$-\frac{1}{2} \iint_{M_0 M_1 M_2} f_1(x, y) dx dy = \frac{1}{2a} \int_0^{at_0} \int_{x_0 - a(t_0 - \tau)}^{x_0 + a(t_0 - \tau)} f(\xi, \tau) d\xi d\tau.$$

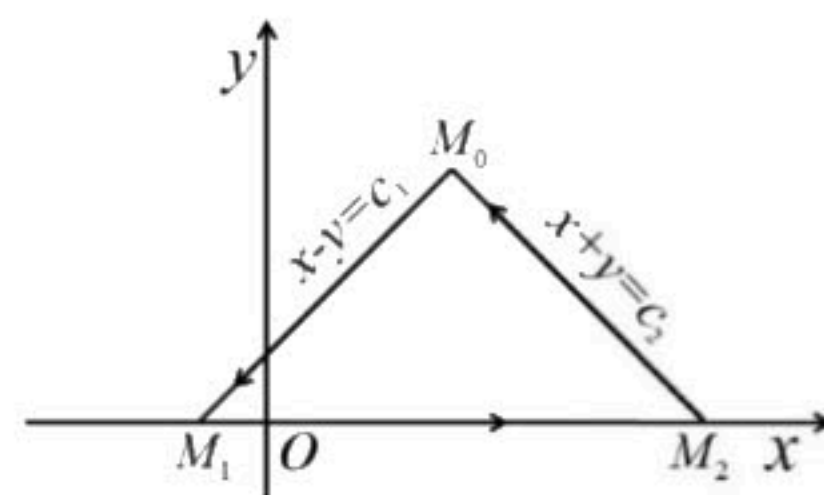


图12-4

故

$$\begin{aligned}
u(x_0, t_0) &= \frac{1}{2} [\varphi(x_0 - at_0) + \varphi(x_0 + at_0)] \\
&\quad + \frac{1}{2a} \int_{x_0 - at_0}^{x_0 + at_0} \psi(\xi) d\xi \\
&\quad + \frac{1}{2a} \int_0^{t_0} \int_{x_0 - a(t_0 - \tau)}^{x_0 + a(t_0 - \tau)} f(\xi, \tau) d\xi d\tau.
\end{aligned}$$

将 x, t 与 x_0, t_0 对调, 并利用黎曼函数的对称性, 即得

$$\begin{aligned}
u(x, t) &= \frac{1}{2} [\varphi(x - at) + \varphi(x + at)] \\
&\quad + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi \\
&\quad + \frac{1}{2a} \int_0^t \int_{x-a(t-\tau)}^{x+a(t-\tau)} f(\xi, \tau) d\xi d\tau.
\end{aligned} \tag{8}$$

当 $f(x, t) = 0$ 时, 原定解问题变成

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, \\ u|_{t=0} = \varphi(x), \\ u_t|_{t=0} = \psi(x). \end{cases}$$

答案⑧退化为

$$u(x, t) = \frac{1}{2} [\varphi(x - at) + \varphi(x + at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi.$$

这正是达朗贝尔公式。

2. 求解 $x^2 u_{tt} - t^2 u_{xx} = 0, u|_{t=1} = \varphi(x), u_t|_{t=1} = \psi(x)$ 。

解：首先，将原问题重写成如下定解问题

$$\begin{cases} x^2 u_{tt} - t^2 u_{xx} = 0, & \textcircled{1} \\ u|_{t=1} = \varphi(x), & \textcircled{2} \\ u_t|_{t=1} = \psi(x). & \textcircled{3} \end{cases}$$

根据教材 (7.3.12) (7.3.13) 判断， $a_{12}^2 - a_{11}a_{22} = x^2 t^2 > 0$ ，故本题方程①为含有两个自变数的变系数双曲型方程。首先，将方程①化成形为

$$\mathbf{L}u \equiv u_{xx} - u_{yy} + b_1 u_x + b_2 u_y + cu = f$$

的标准型。①的特征方程是 $x^2 - t^2 \left(\frac{dx}{dt} \right)^2 = 0$ ，则特征线是 $\ln x \pm \ln t = \text{常数}$ 。作变数变换

$$\begin{cases} y = \ln x \\ z = \ln t \end{cases}$$

则原定解问题变换成

$$\begin{cases} u_{yy} - u_{zz} - u_y + u_z = 0, & \textcircled{4} \\ u|_{z=0} = \varphi(e^y), & \textcircled{5} \\ u_z|_{z=0} = u_t \frac{dt}{dz} \Big|_{t=1} = \psi(e^y). & \textcircled{6} \end{cases}$$

本题中， $a_{11} = 1, a_{22} = -1, b_1 = -1, b_2 = 1, a_{12} = c = 0, \mathbf{M} = \frac{\partial^2}{\partial y^2} - \frac{\partial^2}{\partial z^2} + \frac{\partial}{\partial y} - \frac{\partial}{\partial z}$ ，黎曼函数 v 是 $\mathbf{M}v = 0$ 的解，而且满足

$$\begin{cases} \frac{\partial v}{\partial s} + \frac{b_2 + b_1}{2\sqrt{2}} v = 0 & (\text{在 } M_2 M_0 \text{ 上}), \\ \frac{\partial v}{\partial s} - \frac{b_2 - b_1}{2\sqrt{2}} v = 0 & (\text{在 } M_0 M_1 \text{ 上}), \\ v(x_0, y_0) = 1. \end{cases}$$

从而有

$$\begin{cases} v = e^{\int_{s_0}^s \frac{b_2 + b_1}{2\sqrt{2}} ds} = e^0 = 1, & (\text{在 } M_2 M_0 \text{ 上}), \\ v = e^{\int_{s_0}^s \frac{b_2 - b_1}{2\sqrt{2}} ds} = e^{-(z-z_0)}, & (\text{在 } M_0 M_1 \text{ 上}). \end{cases}$$

取 $v = e^{-\frac{1}{2}(y-y_0+z-z_0)}$ ，

这样，利用推广的格林公式，本问题的一般解是

$$\begin{aligned}
u(y_0, z_0) &= \frac{1}{2} [u(y_1, z_1)v(y_1, z_1) + u(y_2, z_2)v(y_2, z_2)] \\
&\quad + \frac{1}{2} \int_{M_1 M_2} [Y \cos(\mathbf{n}, \mathbf{y}) + Z \cos(\mathbf{n}, \mathbf{z})] ds \\
&\quad - \frac{1}{2} \iint_{M_0 M_1 M_2} v(y, z; y_0, z_0) f(y, z) dS. \quad (7)
\end{aligned}$$

下面分别计算出⑦中各项。由图 12-5 可见, M_1 的坐标为 $y_1 = y_0 - z_0$, $z_1 = 0$, M_2 的坐标为 $y_2 = y_0 + z_0$, $z_2 = 0$ 。则有,

$$\begin{aligned}
u(y_1, z_1)|_{z_1=0} &= \varphi(e^{y_1}) = \varphi(e^{y_0-z_0}) = \varphi\left(\frac{x_0}{t_0}\right), \\
u(y_2, z_2)|_{z_2=0} &= \varphi(e^{y_2}) = \varphi(e^{y_0+z_0}) = \varphi(x_0 t_0), \\
v(y_1, z_1) &= e^{-\frac{1}{2}(y_0-z_0-y_0+0-z_0)} = e^{z_0} = t_0, \\
v(y_2, z_2) &= e^{-\frac{1}{2}(y_0+z_0-y_0+0-z_0)} = e^0 = 1.
\end{aligned}$$

所以, ⑦式右端第一项为

$$\frac{1}{2} [\varphi(e^{y_0-z_0})e^{z_0} + \varphi(e^{y_0+z_0})] = \frac{1}{2} t_0 \varphi\left(\frac{x_0}{t_0}\right) + \frac{1}{2} \varphi(x_0 t_0).$$

由教材(12.5.30)式

$$\begin{cases} Y = -(uv)_y + (2u_y + b_1 u)v = u_y v - uv_y + b_1 uv, \\ Z = (uv)_z - (2u_z - b_2 u)v = uv_z - u_z v + b_2 uv \\ \qquad \qquad \qquad = \frac{1}{2} uv - u_z v, \quad (v_z = -\frac{1}{2} v). \end{cases}$$

从而

$$\begin{aligned}
Z|_{z=0} &= \frac{1}{2} e^{-\frac{1}{2}(y-z_0-y_0)} u|_{z=0} - e^{-\frac{1}{2}(y-z_0-y_0)} u|_{z=0} \\
&= \frac{1}{2} \varphi(e^y) e^{-\frac{y}{2}} \cdot e^{\frac{1}{2}(z_0+y_0)} - \psi(e^y) e^{-\frac{y}{2}} \cdot e^{\frac{1}{2}(z_0+y_0)}.
\end{aligned}$$

$$\cos(\mathbf{n}, \mathbf{y})=0, \quad \cos(\mathbf{n}, \mathbf{z})=-1,$$

所以, $e^y = x$, $e^{-\frac{y}{2}} = x^{-1/2}$, $e^{\frac{1}{2}(z_0+y_0)} = (x_0 t_0)^{1/2}$,

$$Z|_{z=0} = \frac{1}{2} (x_0 t_0)^{1/2} \cdot x^{-1/2} \varphi(x) - (x_0 t_0)^{1/2} \cdot x^{-1/2} \psi(x)$$

又 $dy = \frac{1}{x} dx$, ⑦式右端第二项为

$$\begin{aligned}
\frac{1}{2} \int_{M_1}^{M_2} (-Z) dl &= \frac{1}{2} \int_{y_1}^{y_2} (-Z) dy \\
&= -\frac{\sqrt{x_0 t_0}}{4} \int_{x_0/t_0}^{x_0 t_0} \varphi(x) x^{-3/2} dx + \frac{\sqrt{x_0 t_0}}{2} \int_{x_0/t_0}^{x_0 t_0} \psi(x) x^{-3/2} dx.
\end{aligned}$$

又 $f=0$, 得⑦式右端第三项为零,

$$u(x_0, t_0) = \frac{1}{2} \varphi(x_0 t_0) + \frac{1}{2} t_0 \varphi\left(\frac{x_0}{t_0}\right) - \frac{1}{4} \sqrt{x_0 t_0} \int_{x_0/t_0}^{x_0 t_0} \varphi(\xi) \xi^{-3/2} d\xi + \frac{1}{2} \sqrt{x_0 t_0} \int_{x_0/t_0}^{x_0 t_0} \psi(\xi) \xi^{-3/2} d\xi$$

将 x, t 与 x_0, t_0 对调, 并利用黎曼函数的对称性, 即得

$$u(x, t) = \frac{1}{2} \varphi(xt) + \frac{1}{2} t \varphi\left(\frac{x}{t}\right) - \frac{1}{4} \sqrt{xt} \int_{xt}^{x/t} \varphi(\xi) \xi^{-3/2} d\xi + \frac{1}{2} \sqrt{xt} \int_{xt}^{x/t} \psi(\xi) \xi^{-3/2} d\xi. \quad (8)$$

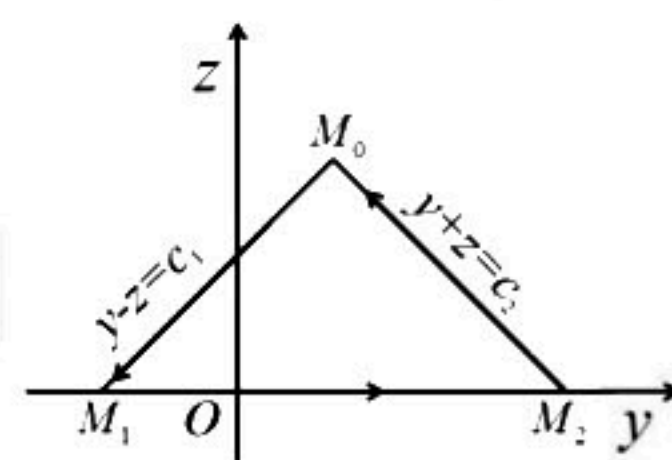


图12-5